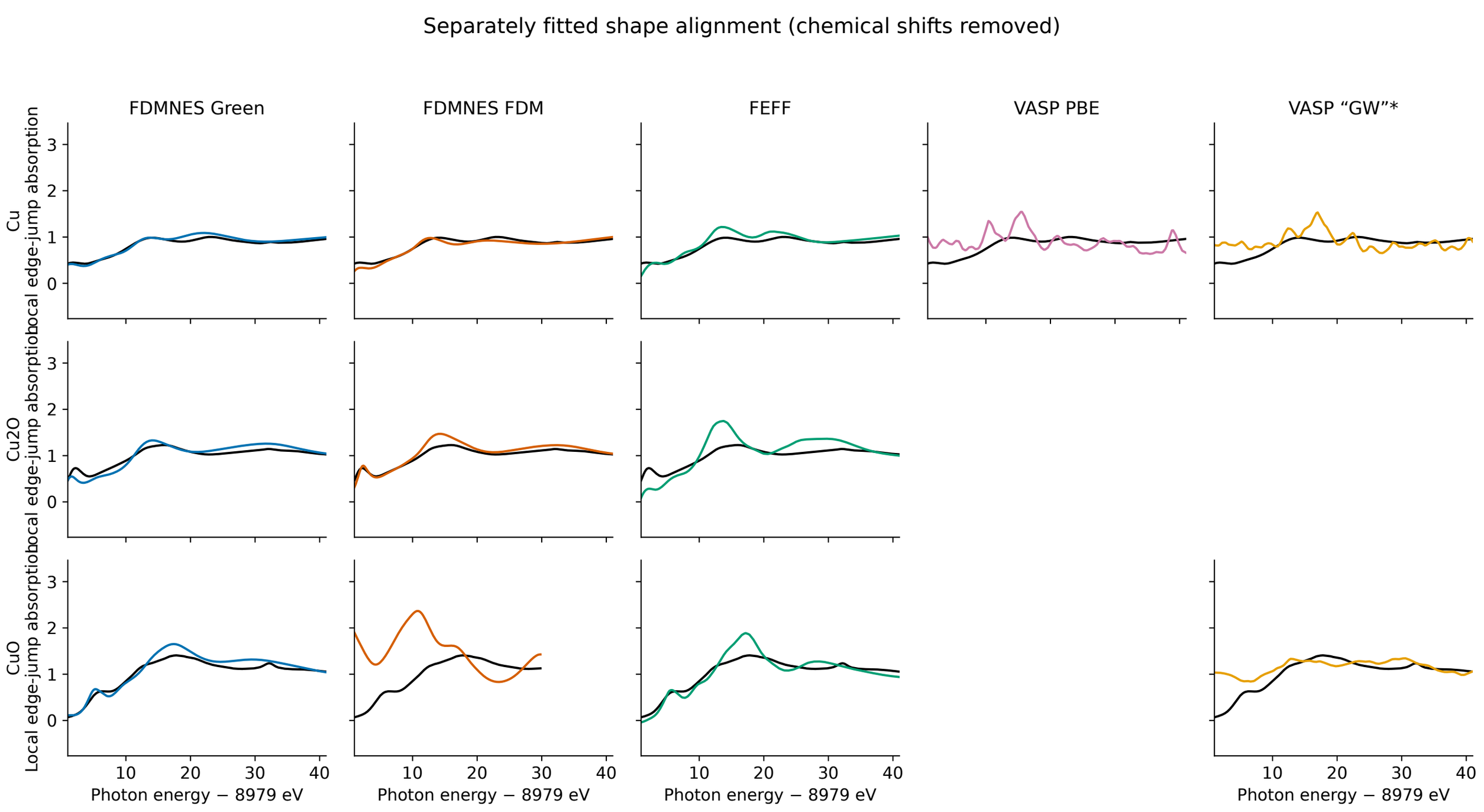
Separately fitted shape alignment (chemical shifts removed)
FDMNES Green
FDMNES FDM
FEFF
VASP PBE
VASP “GW”*
Cu
Cu2O
CuO
Local edge-jump absorption
Photon energy − 8979 eV
Black: supplied experiment. Colors: calculations. *“GW” is a folder alias; no GW/BSE step verified. CuO FDM: core lifetime only, no recorded final-state convolution.